

AGUWA, CHINEDU

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EDUCATION

Bachelor of Engineering (B.Eng.) in Civil Engineering

Federal University of Technology, Minna, Nigeria

Sept 2024

CGPA: 4.79/5.0 (First Class Honours) Top 2 of the department

WORK EXPERIENCE

Environmental Sustainability Software Engineer (Part-time)

EDAT Climate Data and Analytics – Houston, Texas (Remote)

Apr 2025 - Present

- Developed a web application for the complete carbon calculations and accounting. This platform covered a wide range of applications in sustainability.
- Contributed in creating business plan and proposals for the company. Carried out logical calculations of various sustainability areas, closely related to environmental engineering.

Mobile Application & Backend Developer

SureData Consulting LTD – Tilbury (Remote)

Mar 2025 - May 2025

- Develop a mobile application for Church management of activities and members, using react-native, Expo, etc.
- Completed and integrated the backend, cutting across all implementation involved in church operation, using node, express, Postgres DB, ubuntu, etc.

Software Engineering Trainer (Part-time)

A-Z New Age Tutor – Nigeria (Onsite)

Oct 2024 – May 2025

- Delivered practical training session to 5 undergraduate software engineering students covering Python, JavaScript, JavaScript Frameworks, LLMs and machine learning applications.

Mobile Application Developer

Exinn Digital Technology (Start-Up) – Maputo, Maputo Cidade (Remote)

Dec 2024 – Mar 2025

- Developed a medical web application using Typescript, React, Node, Nestjs, Supabase and MongoDB to assist nursing mothers in conceiving.
- Worked on a mobile application for patients using React-native and integrated a chat bot feature for the doctors and a clinical note generator using Gemini AI generative models features.

AI / ML Researcher

Federal University of Technology Minna – Nigeria (Onsite)

Jan 2024 -Sept 2024

- Built an ML model for time-series Bulldozer's sale price prediction and heart diseases using python.
- Developed an AI model for Civil engineering concrete design prediction and integrated in a flutter mobile application.

Freelance Developer

Referrals and Upwork – (remote)

Mar 2020 - Sept 2023

- Built a mobile application for Ride booking with fantastic user experience using React-native.
 - Developed a full-stack Fintech Crypto Application using react-native, expo and coin gecko api.
 - Developed and integrated RESTful APIs, improving tenant management functionality.
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SKILLS & EXPERTISE

- Programming Languages: Python, JavaScript, Go, Typescript, Dart, HTML5, CSS3, SQL
- Libraries/Frameworks: Node.js, Express.js, Nextjs, Firebase, React, Bootstrap, Sass, tailwind, Nestjs
- Databases: MongoDB, MySQL, PostgreSQL
- Mobile Application Development: Flutter/ Dart, React-native
- CI/CD and Project Management Methodologies: Git/GitHub, Expo
- AI / ML: Pandas, MATLAB, NumPy, Scikit-learn, TensorFlow, Jupyter, Excel
- Soft Skills: Teamwork, Effective Communication, Creativity, Research skills, Problem Solving, Leadership, Analytical Thinking, Self-Improvement

RECENT PROJECTS

A Python Model to Predict the Price of Bulldozers

- Developed a time series python model with 88.9% accuracy score, applying Random Forest Regressor machine learning model to predict of the price of bulldozers in the coming year due to inflation.
- Solidified my understanding of regression in supervised learning time series and advanced exploratory Data analysis (EDA).

Development of A Simplified Methodology for British DoE Concrete Mix Design Using Python

- Developed a python-based machine learning predictive model with mean error percentage of 4.3%, to predict the concrete mix design ratio of concrete using the British method.
- Integrated this model into a mobile application to assist civil engineers in performing fast and accurate British Standard Department of environment concrete mix designs while minimizing human error.

Student Assistant Mobile Application (Futmitepadi)

- A mobile app that helps in connecting students at my university. Connecting 210 students at federal university of technology Minna.
- This app has a store where student can sell their old items, has a library for lecture materials, agents for campus accommodation search. Built using Flutter framework, firebase, node, mongo db.

Full Stark Fintech Crypto Website and Mobile Application

- An app that allows tracking of crypto currency prices.
- It was built on React.js, react-native, MySQL, nestjs and coin gecko API.

For more research projects and details, visit [\(Link\)](#).

PUBLICATIONS

- **Aguwa C.** and Abubakar M, (2025). Development of a simplified methodology for British DoE concrete mix design procedure using python. Nile Journal of Engineering and Applied Science. <https://doi.org/10.5455/njeas.238594> [\(Link\)](#)

PROFESSIONAL MEMBERSHIPS

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| • Graduate member, Nigerian Society of Engineers (NSE) | 2024 - Present |
| • Affiliate graduate member, Nigerian institution of civil Engineers (NICE) | 2024 - Present |
| • Member, Young Engineers Forum of Nigeria (YEFoN) | 2024 - Present |
| • Member, International Association of Engineers (Link) (IAENG) | 2024 - Present |

SOME CERTIFICATIONS AND ONLINE COURSES

- Master class on data science using Python
- Introduction to generative AI
- Python Data Structures and Algorithms

LEADERSHIP & VOLUNTEER EXPERIENCE

Executive President (NICESA)

Nigerian Institution of Civil Engineers Student Affiliate (NICESA), FUT Minna 2022 - 2023

- Donated chairs and a sign post to the department laboratory to assist research students. Introduced a yearly mini-scholarship program for two outstanding students (Mr and Miss NICE). Received an award of outstanding executive president due to outstanding leadership. [\(Link\)](#)

Welfare Director (NICESA)

Nigerian Institution of Civil Engineers Student Affiliate (NICESA), FUT Minna 2021 - 2022

- Initiated and facilitate weekly tutorial session for over 250+ civil engineering student and also pre-test to help them prepare properly for examinations. [\(Link\)](#)

CONFERENCES AND PROFESSIONAL TRAINING

- 1st civil engineering seminar: Civil engineering in the 21st century – positioning yourself
Civil engineering department, Federal University of Technology, FUT Minna 2023
- NSE technical session: TVET the lead-way for opportunities, engineering and industrialization
Nigerian Society of Engineers (NSE), Abuja branch 2025